



Shri Yashwantrao Bhonsale Education Society's
YASHWANTRAO BHONSALE INSTITUTE OF TECHNOLOGY

(DTE CODE : 3470) (MSBTE Code : 1742)

Approved by AICTE, DTE & Affiliated to Mumbai University & MSBTE Mumbai
(NBA Accredited ME, CE, EE Diploma Programs)

Practical No 2

Aim

Demonstrate Git & GitHub for Source Code Management

Apparatus / Software Required

- Git
- GitHub Account
- Git Bash or Command Prompt
- Visual Studio Code (VS Code)

Theory

Git: Git is a distributed version control system used to track and manage changes made to source code. It allows developers to maintain different versions of a project and return to previous versions when required.

GitHub: GitHub is a cloud-based platform that hosts Git repositories. It provides features for storing source code, collaboration, code sharing, pull requests, and project management.

Important Git Commands

Command	Purpose
<code>git --version</code>	Checks whether Git is installed
<code>git config --global user.name "Name"</code>	Sets the Git username
<code>git config --global user.email "Email"</code>	Sets the Git email
<code>git init</code>	Initializes a new Git repository
<code>git status</code>	Shows the current status of files
<code>git add .</code>	Adds all files to the staging area
<code>git commit -m "Message"</code>	Saves changes to the local repository
<code>git branch -M main</code>	Renames the current branch to main
<code>git remote add origin URL</code>	Connects the local repository to GitHub
<code>git push -u origin main</code>	Uploads the local repository to GitHub
<code>git pull</code>	Downloads the latest changes from GitHub
<code>git clone URL</code>	Copies a GitHub repository to the local computer



Procedure

Adding a New Project to GitHub Using Git		
Step	Procedure	Command / Action
1	Check Git installation	git --version
2	Configure Git username	git config --global user.name "Your Name"
3	Configure Git email	git config --global user.email "your@email.com"
4	Open your project folder	cd path/to/your/project
5	Initialize Git repository	git init
6	Check repository status	git status
7	Add all project files	git add .
8	Commit the project files	git commit -m "Initial commit"
9	Create a new repository on GitHub	Open GitHub → New Repository → Enter Repository Name → Create Repository
10	Connect local repository to GitHub	git remote add origin <repository-url>
11	Rename branch to main	git branch -M main
12	Upload project to GitHub	git push -u origin main

Make changes to the project (Modify or add project files)		
14	Add updated files	git add .
15	Commit updated files	git commit -m "Updated project"
16	Push updated files to GitHub	git push origin main

Check connected GitHub repository	git remote -v
Download an existing GitHub repository	git clone <repository-url>



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Conclusion:
